STHORAM BMS
ZeroPoint Grid Architecture

Technical White Paper — Volume I

Cell Balancing in $LiFePO_4$ Battery Strings:

The Electrochemical Consequences of Imbalance and the Engineering Case for Zero-Differential Equilibrium

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Abstract

Research published in the *Journal of Power Sources* established that the loss of cyclable lithium inventory — consumed by growth of the solid-electrolyte interphase and associated side reactions — is the primary capacity fade mechanism in $LiFePO_4$ (LFP) cells [Ref 02], a finding confirmed across multiple independent LFP studies [Ref 10, Ref 11]. In a series-connected battery string, cell-to-cell imbalance is the operating condition that forces individual cells outside their design electrochemical window — into overcharge at the top of the string's charge cycle and over-discharge at the bottom — thereby activating and accelerating exactly this degradation mechanism. The engineering inference that connects imbalance to accelerated fade is stated in this paper as such: it follows from the mechanisms documented in the cited literature, and it is the design rationale of the Sthoram BMS architecture.

A second consequence of sustained imbalance is documented in the materials literature: overcharge drives concentration polarization at the anode surface, creating the electrochemical preconditions for lithium dendrite nucleation. A review published in the *Journal of Energy Storage* (2025) identifies concentration polarization caused by the overcharge state as the primary driving force for dendrite initiation, opening the door to metallic lithium deposition on the anode [Ref 01]. The thermal consequence of such deposition has been quantified experimentally: research published in *ACS Applied Materials & Interfaces* (2019) showed that in commercial high-energy graphite-anode lithium-ion cells, lithium plating induced by fast charging reduced the thermal runaway trigger temperature from 215.5°C (cells charged at C/3) to 103.9°C [Ref 12a]. That study was not performed on LFP cells, and this paper does not claim that it was; it is cited because the triggering mechanism it identifies — the exothermic reaction between plated metallic lithium and electrolyte at the graphite anode — is common to graphite-anode cells generally, the LFP/graphite architecture included. Wherever metallic lithium is deposited on a graphite anode, the thermal stability of that cell is degraded relative to its unplated state [Ref 12, Ref 12a].

A third consequence follows from the first two: cells aging at different rates diverge progressively in internal resistance and capacity, compounding imbalance with each cycle. A Monte Carlo simulation study published in the *Journal of Energy Storage* (2020) found that intrinsic variation in cell aging rates is the dominant source of voltage drift across battery pack lifetime — exceeding the influence of initial capacity or impedance variance [Ref 06]. In any series string, pack-level termination is governed by the weakest cell, regardless of the remaining capacity of the others [Ref 14].

Against this evidence base, the balancing behaviour of existing systems is a matter of documented design practice: commercial balancing systems — passive and active — operate to a configurable threshold and stop when that threshold is met. Manufacturer and industry documentation places typical passive thresholds at 30–50 mV [Ref 21], typical active-balancing trigger thresholds at 20–30 mV [Ref 23], and premium configurable systems at 3–5 mV [Ref 24]; a widely deployed automotive BMS documents 10 mV as its recommended maximum delta for stable control [Ref 25]. By way of a recent academic illustration of how demanding equalisation is in practice, a 2025 prototype study of a modular active-balancing BMS reduced a deliberately large initial differential of 1.18 V to approximately 0.47 V over a 2.5-hour balancing session [Ref 26]. None of the systems documented in

the literature surveyed for this paper specifies, as its terminal condition, a zero measured inter-cell differential sustained across all operating states.

The Sthoram BMS Auto-Balancing architecture is designed to achieve complete cell equilibrium as its terminal condition – targeting zero measured inter-cell differential across both operating states – but "zero" here is defined by the limits of the measuring instrument, not absolute mathematical zero. The balancing specification covers both operating states:

1. Idle state: **0 mV** measured inter-cell differential
2. Built-in charging: **0 mV** measured inter-cell differential at charge completion

This paper presents the independent scientific evidence for why imbalance matters, states plainly where the argument rests on Sthoram engineering rationale rather than on a published comparison, and describes a physical 24S $LiFePO_4$ Proof of Concept system on which every specification claimed here can be verified by any qualified party.

What this means in practice: *The research literature establishes the problem: cell imbalance activates the mechanisms by which batteries fade early, fail less safely, and underperform their specification. Commercial balancers stop at a threshold. Sthoram BMS is specified to stop at zero measured differential — and a physical 24-cell system exists on which anyone can check that with standard instruments.*

1. Introduction

Lithium Iron Phosphate ($LiFePO_4$, hereafter LFP) chemistry was selected for the Sthoram BMS system for its thermal stability, the structural robustness of the olivine phosphate framework, and its resistance to the exothermic decomposition pathways that affect layered-oxide chemistries under overcharge. Industry technical documentation reports that the strong P-O bond in LFP releases approximately 70% less oxygen during decomposition than NMC or LCO chemistries [Ref 13 — industry documentation], and comparative thermal-abuse studies consistently place the intrinsic thermal runaway onset of healthy LFP cells far above that of layered-oxide cells [Ref 12].

However, these inherent material advantages are conditional: they hold when individual cells operate within their design electrochemical window. The cathode is not the only component that matters. LFP cells use the same graphite anode as other lithium-ion chemistries, and experimental research on graphite-anode cells has demonstrated that once metallic lithium is plated on that anode, the cell's thermal runaway trigger temperature collapses — from 215.5°C to 103.9°C in the cited fast-charging study on commercial high-energy cells [Ref 12a]. Reviews of LFP degradation identify lithium plating on the graphite anode as a recognised failure mode of the LFP/graphite system as well [Ref 12]. The LFP advantage is real. It is anchored at the cathode — and it can be undermined at the anode by the deposition processes that overcharge sets in motion.

> ***What this means in practice:*** *Think of LFP's safety advantage like a fire-resistant wall. Lithium plating is a crack in that wall, and overcharge of an imbalanced cell is one of the recognised ways the crack starts. It does not repair itself. The role of balancing is to prevent the operating condition that opens the crack in the first place.*

This paper serves two distinct purposes. First, it presents what the scientific literature has independently established about cell imbalance in lithium-ion battery strings – its origins, its electrochemical mechanisms, and its consequences for battery longevity and safety. Second, it states the engineering rationale by which Sthoram BMS concludes that the correct balancing target is zero measured inter-cell differential, rather than any non-zero threshold. These two elements are kept explicitly separate throughout: every statement drawn from published research is cited to its source, and every statement that represents Sthoram's own inference or design reasoning is identified as such.

2. Cell Imbalance in a Series String: Origins and Progression

2.1 Manufacturing Variance

No two lithium-ion cells are identical. Even cells from the same production batch, produced within the same hour on the same equipment, exhibit statistically distributed differences in capacity, internal resistance, and self-discharge rate. This is a fundamental consequence of the stochastic nature of electrode coating, electrolyte impregnation, and formation cycling. Industry engineering documentation further reports that cells from matched production batches can show meaningful capacity variance emerging after 800 to 1,000 cycles — variance not apparent from initial screening [Ref 08 — industry documentation].

MIT RESEARCH CORROBORATION [Ref 30]: Cell-to-cell variance has roots deeper than the production line. A

joint MIT–Stanford study published in Nature (2023) used operando X-ray video imaging of individual carbon-coated $LiFePO_4$ nanoparticles to show that reaction kinetics are measurably heterogeneous even at the single-nanoparticle level **[Ref 30]**. Non-uniformity is intrinsic to the LFP material system at every scale. The inference drawn here is stated as Sthoram engineering rationale: equality between cells is not a natural condition that packs settle into — it is a condition that must be actively engineered and continuously maintained, and the pack level is the only level at which the system designer holds that control.

Peer-reviewed work quantifies the consequence. *ACS Omega* (2023) documented the impact of individual cell parameter differences on series-parallel battery pack performance, finding that in series-connected strings, capacity difference between cells is the dominant factor affecting the pack energy index, and that both capacity difference and ohmic resistance difference significantly affect the pack power index **[Ref 14]**. The critical observation is the asymmetry — the "barrel effect": the weakest cell limits the usable output of the entire pack, while the strongest cell determines the voltage at which charging must terminate.

MIT RESEARCH CORROBORATION [Ref 29]: The scale of real-world variability among nominally identical commercial LFP cells was documented in a landmark MIT–Stanford dataset published in Nature Energy (2019): 124 commercial $LiFePO_4$/graphite cells, cycled under a range of fast-charging conditions, exhibited cycle lives spanning 150 to 2,300 cycles — more than an order of magnitude. The published spread reflects the combined influence of operating conditions and intrinsic device variability **[Ref 29]**. Every series string assembled from commercial cells inherits this variability at the moment of assembly — an observation consistent with the manufacturing-variance evidence above.

What this means in practice: *Imagine a chain of water buckets all connected together. You can only fill them until the smallest bucket overflows — and you can only pour from them until the emptiest one runs dry. The smallest and emptiest bucket controls everything. If those buckets are never made equal, the problem grows worse every time you fill and empty them.*

2.2 Differential Self-Discharge

In an unmanaged series string at rest, cells lose charge at different rates. Self-discharge is driven by internal micro-short-circuit paths, temperature-dependent parasitic reactions at the electrode-electrolyte interface, and electrolyte decomposition reactions that proceed continuously even without external current. These rates vary cell to cell.

Published simulation work has quantified the thermal contribution to this effect. A modelled three-cell string with a 4°C operating temperature differential between cells showed less than 1 mV voltage divergence over 30 days; the same simulation with a 20°C differential — representative of the thermal gradient across a battery cabinet, or between cells adjacent to a heatsink and those further away — produced over 5 mV of divergence [Ref 07 — simulation study]. In a 24S string of cells each subject to individual micro-variations, divergence of this kind accumulates with time at rest and with every cycle.

What this means in practice: *Even when a battery is just sitting unused, the cells inside are slowly drifting*

apart in charge level, and the warmer cells drift faster. By the time the battery is used again, cells that looked equal before have quietly become unequal. Most BMS designs do not correct this during rest. Sthoram BMS does.

2.3 Differential Aging Rates and the Compounding Loop

Imbalance compounds over time. A simulation study published in the *Journal of Energy Storage* (2020), applying Monte Carlo methods to commercially characterised 18650 cells in a large 168S 20P pack model, found that intrinsic variation in cell aging rates is the dominant source of voltage drift across the lifetime of the pack — exceeding the influence of initial capacity or impedance variance [Ref 06 — simulation study].

The mechanism is a positive feedback loop: cells repeatedly stressed at the voltage extremes age faster, which raises their internal resistance, which produces higher local overpotential on the next charge cycle, which increases that stress further. Imbalance left uncorrected does not stay constant. It accelerates.

What this means in practice: *It is like a small wobble in a wheel that gets worse the faster you drive. The wobble causes wear, the wear makes the wobble worse, and eventually the wheel fails — long before the car should have needed new wheels. The only fix is to remove the wobble at the source, on every cycle.*

2.4 The 4.5% Imbalance Experiment: What One Charge-Discharge Cycle Does

A controlled experiment published in *Energies* (MDPI, 2021) studied a $LiFePO_4$ battery pack in which an initial SOC imbalance of only 4.5% was introduced between cells. The result was immediate and unambiguous: the weakest cell was driven into over-discharge — below the lower allowable voltage limit of 2 V — on the very first discharge cycle, with reverse polarity observed in the final phase. The imbalance did not self-correct over subsequent cycles [Ref 15].

KEY FINDING [Ref 15]	A 4.5% initial SOC imbalance in a $LiFePO_4$ pack was sufficient, in this controlled experiment, to drive a cell into over-discharge with reverse polarity on the very first cycle. The imbalance did not self-correct in the absence of active balancing: it was present on every subsequent cycle.

What this means in practice: A 4.5% imbalance — which sounds negligible — was enough to cause immediate, self-repeating damage from the very first discharge. This is not an edge case or a worst-case scenario. Imbalance is not a minor inconvenience you can tolerate now and correct later. It is a built-in structural flaw that begins causing damage from cycle one and never stops.

3. The Electrochemical Consequences of Imbalance

3.1 Concentration Polarization and the Overpotential Threshold

In a series-connected battery string, every cell carries the same current — a fundamental constraint of series topology. When a cell with lower capacity or higher internal resistance approaches its upper

voltage limit while others continue to receive charge current, the local overpotential at its anode increases.

A review published in the *Journal of Energy Storage* (2025) identifies concentration polarization — driven by the overcharge state — as the primary driving force initiating lithium dendrite growth. Overcharge produces a sharp imbalance in the local Li^+ concentration at the negative electrode surface; when the local deviation from equilibrium potential exceeds the nucleation threshold for metallic lithium deposition, Li^+ ions that can no longer intercalate into fully lithiated graphite begin to electrodeposit as metallic lithium on the anode surface [Ref 01].

> ***What this means in practice:*** *When a cell is full but the charger keeps pushing current into it, the lithium ions run out of places to go inside the graphite. So instead of going in, they pile up on the surface and start forming metallic deposits. This is exactly the condition cell balancing exists to prevent: when all cells are equal, no cell is ever full while others are still charging.*

3.2 Lithium Plating and Dendrite Nucleation

The Barton-Bockris model, established in 1962 and consistently supported by subsequent experimental and computational research, describes why electrodeposited lithium develops dendritic morphology. Deposition at electrode surface protrusions — surface inhomogeneities, SEI defects, grain boundaries — is geometrically preferred because local Li^+ diffusion flux is enhanced at protruding sites; preferential deposition amplifies existing surface features, causing deposits to elongate into the electrolyte toward the cathode [Ref 09].

In a balanced cell operating within its electrochemical window, this mechanism is suppressed because the anode potential does not approach the Li^+/Li^0 deposition threshold. The intercalation reaction proceeds normally: $Li^+ + e^- + C_6 \rightarrow LiC_6$. In a cell subjected to overcharge, fully lithiated graphite cannot accept further Li^+ and the reaction shifts toward electrodeposition: $Li^+ + e^- \rightarrow Li^0$. The distinction between these two reactions is the difference between a healthy cell and a cell on a path toward internal short circuit.

> ***What this means in practice: In a balanced cell, lithium ions slot neatly into graphite — like keys going into locks. In an overcharged cell, all the locks are already full, so the keys have nowhere to go. Instead of entering, they pile up on the surface and form metallic deposits. Those deposits can grow inward and eventually pierce the separator. Balancing ensures no cell is ever forced to keep receiving lithium when it is already full — so the deposits never form in the first place.***

3.3 SEI Disruption and Irreversible Active Lithium Loss

The solid-electrolyte interphase (SEI) is a thin passivation layer that forms on the graphite anode during initial charge-discharge cycles. When stable, it conducts Li^+ while blocking electron transfer, preventing continued electrolyte decomposition. Its stability is contingent on the electrode remaining within its design electrochemical window.

Plated metallic lithium reacts readily with electrolyte components, consuming both active lithium and electrolyte irreversibly. A representative reaction involving ethylene carbonate (EC), a primary electrolyte solvent, produces lithium carbonate and ethylene gas:

$$2\ Li^0 + C_3H_4O_3\ (EC) \rightarrow Li_2CO_3 + C_2H_4$$

The lithium carbonate deposits over the existing SEI, producing a thicker, less uniform, higher-resistance interphase. Fundamental SEI research in *Batteries & Supercaps* (Wiley, 2021) documents how SEI formation consumes lithium and electrolyte, with growth accelerating at higher current densities **[Ref 04]**, and an operando electron-microscopy study (2023) established that SEI formation is a continuous, mosaic-like evolution that consumes active lithium inventory throughout cell lifetime **[Ref 05]**. Loss of active lithium inventory (LLI) through SEI growth has been confirmed across multiple independent LFP studies as the primary capacity fade mechanism [Ref 02, Ref 10, Ref 11].

> ***What this means in practice:*** *Every time a cell is pushed outside its window, some of the lithium inside it gets permanently locked away in chemical reactions and can never be used again. The battery's capacity drops permanently. There is no way to get that lithium back. Balancing prevents this by preventing the overcharge that triggers it.*

3.4 The Thermal Safety Consequence

The most critical safety consequence of lithium plating on a graphite anode is the collapse of thermal stability. Research published in *ACS Applied Materials & Interfaces* (2019) quantified this directly on commercial high-energy-density graphite-anode lithium-ion cells: cells that experienced lithium plating as a result of fast charging exhibited a thermal runaway trigger temperature (T_2) of 103.9°C, whereas cells charged at a controlled low rate (C/3) exhibited T_2 of 215.5°C **[Ref 12a]**. The study identified the exothermic reaction between plated lithium and electrolyte as the triggering mechanism of the earlier runaway.

Two points of scope must be stated precisely. First, the cited experiment induced plating through fast charging, not through string imbalance; it is cited here because overcharge of an imbalanced cell produces metallic lithium deposition on the graphite anode through the same electrode process (Section 3.1–3.2, [Ref 01]), so the thermal consequence it measured is the consequence of the deposit, however produced. Second, the study's cells were high-energy-density cells, not LFP; the specific temperature values should not be read as LFP/graphite numbers. What the study establishes — and what reviews of LFP degradation corroborate for the LFP/graphite system [Ref 12, Ref 18] — is that plated lithium on a graphite anode drastically lowers the temperature at which thermal runaway can be triggered, in any cell that carries it.

KEY FINDING [Ref 12a]	In commercial graphite-anode lithium-ion cells, lithium plating reduced the measured thermal runaway trigger temperature from 215.5°C to 103.9°C — a reduction of 111.6°C, roughly half the thermal margin — with the plated-lithium/electrolyte reaction identified as the trigger. The study concerned fast-charging-induced plating in high-energy cells; the mechanism by which overcharge in an imbalanced string deposits lithium on graphite is the same electrode process [Ref 01], which is why preventing overcharge is a thermal-safety measure and not only a longevity measure.

What this means in practice: *A battery's famous heat tolerance assumes a clean anode. Once metallic lithium is plated on the graphite, the temperature at which the cell can run away drops dramatically — in the cited study, to around 104°C, a temperature reachable in a fire, a hot enclosure, or a failing neighbouring cell. Balancing protects the condition that keeps the anode clean.*

4. The State of the Art: Threshold-Terminated Balancing

Before presenting what complete balance produces, it is necessary to establish a factual baseline about how existing balancing systems terminate. Across the technical literature and manufacturer documentation surveyed for this paper, balancing systems — passive and active — are specified to operate to a configurable voltage threshold and to stop when that threshold is met. We are not aware of any commercially available BMS whose published specification defines its terminal condition as zero measured inter-cell differential maintained across all operating states; readers are invited to test this statement against any system documentation of their choosing.

4.1 Passive Balancing: Operates to a Threshold, Not to Zero

Passive balancing — the most widely deployed method globally — dissipates excess energy from higher-voltage cells through bypass resistors until cell voltages fall within a pre-set threshold of each other. The threshold at which balancing is declared complete is a design parameter, not a physical limit. Industry documentation places typical passive thresholds at 30 to 50 mV [Ref 21 — industry documentation]; high-quality passive systems may operate at 10 mV.

Once the differential falls below the threshold, the bleeding stops — not because the cells are equal, but because the control algorithm has decided the remaining difference is acceptable. The residual differential is by design, not by accident.

MANUFACTURER DOCUMENTATION [Ref 22]	EM Battery Systems technical documentation states that “balancing to ±1 mV is not practical” for passive balancing systems. This is a vendor engineering statement rather than a peer-reviewed finding, but it is consistent with the threshold-terminated design of passive balancers documented across the industry [Ref 21, Ref 23, Ref 24, Ref 25].

What this means in practice: *Passive balancing is like pouring water out of the fullest buckets until they are roughly the same level. You never get them perfectly equal — you stop when they are close enough. The water you poured out is gone forever. And the cells are never truly equal.*

4.2 Active Balancing: Faster, But Still Threshold-Terminated

Active balancing systems transfer energy between cells using capacitors, inductors, or DC-DC converters rather than dissipating it. This is more efficient than passive balancing and achieves better SOC uniformity. However, the active systems documented in manufacturer specifications share the same terminal logic: they trigger above a configurable threshold and cease when the differential is resolved to it. DALY documents default trigger thresholds of 20 to 30 mV for its active balancers [Ref 23]; premium configurable systems reach 3 to 5 mV [Ref 24]; and the Orion BMS — a widely used

automotive BMS — documents 10 mV as its recommended maximum delta for LFP, noting that lower settings cause control instability in its balancing algorithm [Ref 25].

Academic prototype work illustrates how demanding equalisation is in practice. A 2025 study in *Batteries* (MDPI) of a modular BMS with DC-DC-converter active balancing reported reducing a deliberately large initial voltage difference of 1.18 V to approximately 0.47 V over a 2-hour 30-minute balancing operation on a 22-cell prototype **[Ref 26]**. This figure is cited as a worked example of active-balancing dynamics from a large initial imbalance — not as a performance ceiling of the field, and not as a representation of what commercial systems achieve from ordinary starting conditions.

> ***What this means in practice:*** *Active balancing is smarter and faster than passive balancing. But by design it still stops at "close enough" — a threshold — rather than at "equal." Sthoram BMS is specified to stop at equal.*

4.3 Why Zero Has Not Been the Terminal Condition

The reason existing systems do not specify zero is structural, not a lack of effort. All architectures documented above share a common constraint: the balancing circuit is separate from the charge path, operates through active electronics (resistors, MOSFETs, inductors, capacitors, DC-DC converters), and terminates on a control threshold. Every control-threshold system has a design floor below which it does not act — and that floor is never zero. There is also a measurement dimension: engineering literature on high-voltage BMS reports that real-world voltage measurement error in commercial BMS hardware is constrained to roughly the 5 mV level under field conditions, even on systems with 1 mV nominal resolution, due to calibration drift, thermal effects and sense-wire noise **[Ref 27]**. A system that cannot reliably resolve differences below ~5 mV cannot reliably regulate to below ~5 mV — and, equally, any claim of sub-millivolt balance must be verified with instrumentation better than the claim, a point addressed directly in the verification protocol of Section 7.

THE DESIGN RATIONALE	Any tolerated residual differential is the starting error of the next cycle, and Section 2.3 documents that cell-to-cell divergence is self-reinforcing. To our knowledge, no published study has yet directly compared multi-thousand-cycle outcomes of 0 mV balancing against small non-zero thresholds — such a system has not existed to test. The inference that the only terminal condition introducing zero compounding error is zero differential is therefore stated as Sthoram engineering rationale, grounded in the degradation mechanisms of Sections 2-3, and it is exactly the proposition the Proof of Concept system exists to let others examine.

> ***What this means in practice:*** *If you are slightly off target every time you fire, the error seems tiny — but across 4,000 shots it defines everything. The same logic drives this design: remove the residual entirely, every cycle, so there is nothing left to compound. That is a design philosophy, honestly labelled as such — and it is testable on real hardware.*

5. What Complete Cell Balance Produces: Evidence from Research

5.1 Elimination of the Overcharge Pathway

The most fundamental consequence of maintaining complete voltage balance across all cells in a series string is structural: the conditions under which lithium plating initiates are not met. When every cell begins each charge event at the same voltage, carries the same SOC, and reaches the upper voltage limit simultaneously, no cell is forced to continue receiving charge current after reaching full capacity. The concentration-polarization mechanism that drives dendrite nucleation [Ref 01] requires a cell to be pushed past its capacity limit while the string continues charging; when balance is exact, that condition cannot arise.

> ***What this means in practice:*** *Perfect balance means no cell ever gets more charge than it can hold while others are still filling up. The cause-chain for dendrite formation — which starts with overcharge — is removed before it can begin.*

5.2 SEI Stability and Reduced Active Lithium Loss

When the anode potential is held within the graphite intercalation window — as it is when all cells operate in balance — the SEI is not disrupted by metallic lithium deposition. The practical consequence is reduced LLI: a balanced cell that never experiences overcharge accumulates LLI only through slow calendar-aging reactions, not through the additional cycle-driven mechanism of plated-lithium/electrolyte reaction. The LFP literature consistently identifies LLI as the primary fade mechanism and documents its acceleration under out-of-window operation [Ref 02, Ref 10, Ref 11].

> ***What this means in practice: In a balanced system, the protective layer on each cell stays intact and stable. No active lithium is consumed beyond normal calendar aging — meaning the battery retains its full capacity cycle after cycle. In an imbalanced system, that layer is repeatedly damaged and rebuilt, consuming active lithium each time.***

5.3 Uniform Depth of Discharge and Equal Electrode Wear

In a balanced series string, every cell begins each discharge at the same SOC and reaches the lower voltage cutoff simultaneously. Every cell delivers the same fraction of its capacity on every cycle; DOD is equal across all 24 cells; no cell is over-discharged; no cell cycles over a narrower or wider SOC window than any other. Electrode wear is distributed uniformly, and the differential-aging feedback loop described in Section 2.3 is denied its starting condition. Over-discharge is documented to decompose organic SEI components and accelerate LLI [Ref 10], and inhomogeneous lithium content from cycling over different SOC windows is documented to cause persistent capacity loss in LFP cells [Ref 12].

> ***What this means in practice:*** *All 24 cells age at the same rate because they all do the same work on every cycle. No cell gets worn out faster than its neighbours — so the pack lasts as long as its cells, not as long as its weakest cell.*

5.4 Pack Capacity Utilisation

In a balanced string, all cells reach the charge-termination voltage simultaneously — no cell is under-charged — and all cells reach the discharge cutoff simultaneously — no energy is stranded in

higher-SOC cells. The full installed capacity of the pack is accessible on every cycle. *ACS Omega* (2023) confirmed that capacity difference between cells is the dominant factor limiting pack energy utilisation in series strings [Ref 14].

> ***What this means in practice:*** *A balanced battery gives you the full energy you paid for on every charge-discharge cycle. An imbalanced battery gives you less — because the pack stops when the weakest cell runs out, even if the others still have charge left.*

5.5 Quantified Research Outcomes

The table below reproduces figures exactly as published, with the nature and scope of each source stated in the first column.

Research Finding (source type)	Imbalanced / Degraded Condition	Balanced / Corrected
Thermal runaway trigger temperature, commercial graphite-anode high-energy Li-ion cells; plating induced by fast charging (experimental — ACS Appl. Mater. Interfaces, 2019) [Ref 12a]	103.9°C (Li plating present)	215.5°C (no plating; C/3 charging)
Cycle life impact of 20% internal-resistance variance between parallel-connected cells (figure reported in a survey preprint, arXiv 2024) [Ref 17]	~40% cycle life reduction	Baseline life maintained
Cycle life of SOH-based balancing vs. conventional active SOC balancing (simulation case study — Frontiers in Energy Research, 2022) [Ref 16]	Conventional SOC balancing – baseline	+52.9% cycle life (simulated)
Pack lifespan reduction attributed to imbalance (industry documentation, 2025) [Ref 08]	Up to 30% pack lifespan reduction	Full design lifespan
4.5% initial SOC imbalance in an LFP pack (experimental – Energies, MDPI, 2021) [Ref 15]	Over-discharge on first cycle; reverse polarity observed	Normal operation across all cells

6. Cell Balancing Across Both Operating States

The electrochemical risks from imbalance are not confined to a single operating mode. They are present, in different forms, across the operating states addressed here: Idle State and Built-in charging. The Sthoram BMS specification addresses each state explicitly and with a distinct balancing response.

6.1 Idle State — Target: 0 mV Measured Differential

The idle state is electrochemically consequential because it is when self-discharge divergence accumulates unimpeded (Section 2.2). It is also where honest treatment of LFP's voltage-SOC characteristic matters most. LFP exhibits a famously flat open-circuit-voltage plateau between roughly 20% and 80% SOC; industry measurement documentation notes that a difference of a few tens of millivolts at rest can mask a very large SOC difference on this plateau [Ref 28]. Two consequences follow, and the Sthoram specification is written around both. First, voltage equality at rest on the plateau is a necessary but not by itself sufficient indicator of SOC equality — which is precisely why the specification does not rely on idle-state voltage alone: full SOC equalisation is

completed during built-in charging (Section 6.2), where all cells are taken together to the top of charge, the region in which the voltage-SOC relationship is steep and voltage equality and SOC equality converge. Second, because small voltage differences on the plateau can correspond to large SOC differences, tolerating a residual differential at rest is more consequential for LFP than the millivolt figure suggests — the basis of the **0 mV** idle target.

> ***What this means in practice:*** *After the battery has been sitting unused, all 24 cells are brought to the same measured voltage before it is used again — and true charge-level equality is locked in at the next built-in charge, where the cell voltages become sensitive again. Every new cycle starts from a clean slate.*

6.2 Built-In Charging — Target: 0 mV at Completion

During built-in charging, the Sthoram BMS reconfigures the 24S string so that all 24 cells charge simultaneously at uniform voltage — not sequentially through a series topology. Because every cell receives the same charging voltage at the same time, every cell reaches the upper voltage limit at the same moment; no cell is forced to continue receiving charge current after reaching full capacity; the overcharge pathway is structurally closed; and the inter-cell differential converges to zero measured difference at completion.

In conventional series charging, by contrast, the cell with the lowest capacity reaches full charge first and is stressed while the string continues charging. Peer-reviewed overcharge studies document the degradation cascade this produces in LFP cells **[Ref 11, Ref 18]**, and industry technical documentation reports representative figures of 15–30% internal-resistance growth over 100 cycles and capacity retention falling to 80% within 200 cycles under chronic mild overcharge [Ref 13 — industry documentation]. The Sthoram built-in charging architecture eliminates this condition by design.

> ***What this means in practice:*** *With the built-in charger, all 24 cells charge together, side by side, at the same voltage. They all finish at the same moment, all equally full. There is no last cell to finish while the others wait in overcharge.*

7. Proof of Concept: Independent Verification Protocol

The specifications stated in this document are not offered as theoretical projections to be taken on trust. They are implemented on a physical 24S $LiFePO_4$ battery pack — the Sthoram BMS Proof of Concept system — which is available for independent verification by qualified engineers, research institutions, or accredited testing bodies. Any party may connect their own data-logging equipment or voltmeter and observe the two-claim verification protocol below in real time.

A note on measurement, stated up front because Section 4.3 requires it: field-deployed BMS measurement chains are documented to carry roughly 5 mV of real-world uncertainty **[Ref 27]**. A claim of zero differential is therefore only as strong as the instrument that checks it. Accordingly, "**0 mV**" in this protocol means no measurable inter-cell difference at the resolution of the verifying instrument, and independent verifiers are expressly encouraged to bring calibrated laboratory-grade instrumentation of finer resolution than the on-board electronics — the claim is made to be tested at whatever resolution the verifier can bring.

The mechanism by which Sthoram BMS achieves these specifications is the subject of 25 patent applications filed in India (accepted 13 June 2026; international filing eligible from 25 July 2026 under Section 39, Indian Patents Act, 1970). The architecture removes active balancing electronics from the current path, generates no balancing heat, and operates without component wear — properties that follow from the underlying patented mechanism and that distinguish it structurally from the active and passive topologies documented in Section 4. The mechanism is not disclosed in this document. The outcomes it produces are demonstrated on the physical system.

Claim 1 — 0 mV Measured Balance on Built-In Charging

Starting condition: the 24S pack is presented with cells at measurably different voltages — confirmed by data logger or voltmeter before the test begins.

Action: the built-in charger is connected.

Observable: all 24 cells balance during the charge event; cell voltages converge in real time, visible on the data logger or application. **End condition:** on charge completion, all 24 cells read identical voltage — no measurable inter-cell differential at the resolution of the verifying instrument.

CLAIM 1 OPEN TO VERIFICATION	Zero measured inter-cell differential at built-in charging completion — verifiable by any instrument, by any party, on the physical POC system.

Claim 2 — 0 mV Measured Equilibrium in Idle State

Starting condition: the cells are close in voltage but not equal — for example: Cell A 3.213 V, Cell B 3.216 V, Cell C 3.210 V, and so on across all 24 cells.

Action: the load is disconnected; the pack enters the idle state.

Observable: within a maximum of 2 hours, all 24 cells converge to the same measured voltage — for example, all cells arriving at 3.213 V — with no external intervention and no operator action.

CLAIM 2 OPEN TO VERIFICATION	Zero measured differential achieved autonomously in the idle state; all 24 cells converge to identical measured voltage — verifiable by data logger over the convergence period.

Together the two claims constitute a complete, reproducible, independently auditable verification protocol for the Sthoram BMS balancing specification. The POC is a physical system; the instruments are standard; the results are observable by anyone. This is not a simulation and not a model — it is a demonstration, and it is offered for scrutiny.

8. Summary of Documented Outcomes and Their Sources

The findings below are drawn from the sources cited in Section 10. Each entry states the nature of its source. Peer-reviewed findings, simulation results, survey figures and industry documentation are different classes of evidence, and they are labelled accordingly.

- Concentration polarization from the overcharge state is identified as the primary driving force for lithium dendrite initiation. *(Peer-reviewed review, 2025)* **[01]**
- Intrinsic variation in cell aging rates is the dominant source of voltage drift across pack lifetime, exceeding initial capacity or impedance variance. *(Peer-reviewed Monte Carlo simulation study)* **[06]**
- A 20°C thermal differential between cells produced over 5 mV of voltage divergence over 30 days in the absence of balancing. *(Peer-reviewed simulation study)* **[07]**
- Cells from the same production batch can show significant capacity variance after 800–1,000 cycles. *(Industry documentation)* **[08]**
- Passive balancing to ±1 mV is described by a manufacturer as not practical; commercial balancers documented here terminate on thresholds. *(Manufacturer documentation)* [22, 21, 23, 24, 25]
- A 2025 academic prototype reduced a deliberately large 1.18 V inter-cell differential to ~0.47 V over 2.5 hours — a worked example of active-balancing dynamics from a large initial imbalance. *(Peer-reviewed prototype study)* **[26]**
- Lithium plating reduced the thermal runaway trigger temperature of commercial graphite-anode high-energy Li-ion cells from 215.5°C to 103.9°C; the plated-lithium/electrolyte reaction was the trigger. The study induced plating by fast charging and did not use LFP cells; the anode mechanism is common to graphite-anode chemistries. *(Peer-reviewed experimental study)* **[12a]**
- Chronic mild overcharge is reported to raise internal resistance 15–30% over 100 cycles and cut capacity retention to 80% within 200 cycles. *(Industry documentation)* **[13]**
- Capacity difference between cells is the dominant factor limiting pack energy utilisation in series strings. *(Peer-reviewed study)* **[14]**
- A 4.5% initial SOC imbalance drove the weakest LFP cell into over-discharge with reverse polarity on the first cycle. *(Peer-reviewed experiment)* **[15]**

- SOH-optimised balancing produced a 52.9% cycle-life improvement over conventional active SOC balancing in a simulated 3-pack / 15-cell case study. *(Peer-reviewed simulation)* [16]
- A 20% internal-resistance variance between parallel-connected cells is associated with roughly 40% cycle-life reduction. *(Figure reported in a survey preprint; parallel configuration)* [17]
- Over-discharge decomposes organic SEI components, driving destruction-repair cycles that accelerate LLI. *(Peer-reviewed study)* [10]
- Inhomogeneous lithium content from cycling over different SOC windows causes persistent, measurable capacity loss in LFP cells. *(Peer-reviewed review)* [12]
- LLI through SEI growth is the primary capacity fade mechanism in LFP/graphite cells. *(Peer-reviewed studies)* [02, 10, 11]

9. Conclusion

The scientific evidence on cell imbalance in lithium-ion battery strings is consistent across chemistry types, research institutions, and methodologies. Imbalance activates a cascade of electrochemical degradation mechanisms whose consequences are measurable, cumulative, and irreversible. The balancing systems documented in the literature and manufacturer specifications surveyed here — passive and active alike — terminate on a threshold, and that threshold, however small, leaves a residual differential in place at the start of every subsequent cycle.

Whether a residual of a few millivolts materially shortens service life relative to true zero is, today, an open empirical question — no commercial zero-differential system has existed against which to test it. Sthoram BMS takes a clear engineering position on that question: the degradation mechanisms documented in Sections 2 and 3 operate whenever imbalance is present and are suppressed whenever it is not, and the only terminal condition that introduces no compounding error is zero. That position is stated as design rationale, not as a published finding — and, uniquely, it is now testable, because the system built on it exists.

A physical Proof of Concept — a 24S $LiFePO_4$ battery pack — demonstrates both balancing claims with independently verifiable, real-time, instrument-readable results, under the measurement terms stated openly in Section 7. The mechanism that achieves these results is protected by 25 patent applications accepted in India (13 June 2026; international filing eligible from 25 July 2026 under Section 39, Indian Patents Act, 1970). The outcomes those applications protect are documented in this paper and demonstrated on the physical system.

These are not assertions offered for belief. They are demonstrations offered for measurement.

> ***What this means in practice:*** *To the best of our knowledge, no commercially available BMS specifies zero measured inter-cell differential, sustained across both operating states, as its terminal condition. Sthoram BMS does — on a real 24-cell battery pack, measurable by anyone, with standard instruments, at any time. We invite the measurement.*

10. Bibliography

References are classified as follows: peer-reviewed publications and archival preprints are cited for scientific findings; manufacturer and industry documentation is cited only for statements about commercial design practice and is labelled *[industry documentation]* throughout. Reference numbering is preserved from Version 2.0 of this paper for continuity; entry [03] of Version 2.0 (a non-indexed proceedings item) has been withdrawn in this edition, and the Barton-Bockris model is cited from its primary source [09]. Entry [19] of Version 2.0 duplicated entry [12] and has been merged into it. Entry [20] of Version 2.0 was not cited in the body of this edition and has been removed.

[01] Liu, Y.-C., Li, Y., Fu, X., Xue, Z.-A. & Wang, J.-J. (2025). "Lithium dendrite growth, evolution, inhibition in lithium-ion batteries." *Journal of Energy Storage*, 138, 118559. DOI: 10.1016/j.est.2025.118559. [Review; identifies concentration polarization from the overcharge state as the primary driving force for dendrite initiation.]

[02] Zhang, Y., Wang, C.-Y. & Tang, X. (2011). "Cycling degradation of an automotive $LiFePO_4$ lithium-ion battery." *Journal of Power Sources*, 196(3), 1513–1520. DOI: 10.1016/j.jpowsour.2010.08.070. [Single-cell experimental study of a 16.4 Ah prismatic LFP cell; identifies loss of cyclable lithium from SEI growth and side reactions as the primary capacity fade mechanism.]

[04] Nojabaee, M. et al. (2021). "Understanding the Nature of the Solid-Electrolyte Interphase on Lithium Metal in Liquid Electrolytes." *Batteries & Supercaps*, Wiley. DOI: 10.1002/batt.202000258. [Fundamental SEI characterisation; cited for SEI growth behaviour and its lithium/electrolyte consumption.]

[05] Operando EC-LC-STEM Investigation (2023). PMC ID: 10604081. "Growth and Evolution of the Mosaic SEI for Lithium-Ion Batteries." [SEI as a continuous mosaic evolution consuming active lithium throughout lifetime.]

[06] Zilberman, I., Schmitt, J., Ludwig, S., Naumann, M. & Jossen, A. (2020). "Simulation of voltage imbalance in large lithium-ion battery packs influenced by cell-to-cell variations and balancing systems." *Journal of Energy Storage*, 32, 101828. DOI: 10.1016/j.est.2020.101828. [Monte Carlo simulation study; intrinsic aging-rate variation as the dominant source of pack voltage divergence.]

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[08] Large-Battery.com (2025). "Understanding Lithium Battery Cell Imbalances" / "The Role of Cell Balancing in Extending Battery Lifespan." [Industry documentation — not peer-reviewed; capacity variance after 800–1,000 cycles; up-to-30% lifespan figure.]

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[10] Tang, S. et al. (2025). "Revisiting the overdischarge process as a novel accelerated aging method for $LiFePO_4$/Graphite batteries through the unveiling of SEI evolution mechanism." *Energy Storage Materials*, 74, 103916. DOI: 10.1016/j.ensm.2024.103916. [SEI organic-component decomposition under over-discharge; LLI from SEI growth as the predominant SOH decline factor.]

[11] "Experimental Study of Degradation Characteristics of $LiFePO_4$ During Overcharging at Low Temperatures" (2023). *Energies*, 16(6), 2786. MDPI. DOI: 10.3390/en16062786. [Capacity decrease under overcharge; LLI as the primary mechanism in LFP.]

[12] Wang, L. et al. (2022). "Insights for understanding multiscale degradation of $LiFePO_4$ cathodes." *eScience*, 2(2), 125–137. DOI: 10.1016/j.esci.2022.03.006. [Review of LFP degradation, including lithium plating on the graphite anode and persistent capacity loss from inhomogeneous lithium content under partial-SOC cycling. This entry consolidates entries [12] and [19] of Version 2.0, which cited the same publication.]

[12a] Li, Y., Feng, X., Ren, D., Ouyang, M., Lu, L. & Han, X. (2019). "Thermal runaway triggered by plated lithium on the anode after fast charging." *ACS Applied Materials & Interfaces*, 11(50), 46839–46850. DOI: 10.1021/acsami.9b16589.

[Experimental study on commercial high-energy-density graphite-anode lithium-ion cells: thermal runaway trigger temperature of 103.9°C after fast-charging-induced lithium plating versus 215.5°C for cells charged at C/3, with the plated-lithium/electrolyte reaction identified as the trigger. Note: this study was not conducted on LFP cells; it is cited in this paper as evidence of the thermal consequence of lithium plating on graphite anodes generally.]

[13] DEESPAEK Lithium Battery Technical Report (2025). "How to Prevent Overcharging Risks in LFP Batteries." [Industry documentation — not peer-reviewed; reported figures: chronic overcharging at 3.8 V/cell → IR increase 15-30% over 100 cycles; capacity retention 80% within 200 cycles; ~70% lower oxygen release for LFP.]

[14] Wang, Y. et al. (2023). "Impact of Individual Cell Parameter Difference on the Performance of Series-Parallel Battery Packs." *ACS Omega*, 8(11), 10512-10524. DOI: 10.1021/acsomega.3c00266. [Capacity difference as the dominant factor affecting pack energy index in series strings; the "barrel effect."]

[15] Madej, W. & Wojciechowski, A. (2021). "Analysis of the Charging and Discharging Process of $LiFePO_4$ Battery Pack." *Energies*, 14(13), 4055. MDPI. DOI: 10.3390/en14134055. [4.5% initial SOC imbalance drives the weakest LFP cell into over-discharge with reverse polarity on the first cycle.]

[16] Li, X., Yin, X., Tian, Z., Jiang, X., Jiang, L. & Smith, J. (2022). "Multi-layer state of health balancing control for a battery-based energy storage system to extend cycle life based on active equalization circuits." *Frontiers in Energy Research*, 10:966422. DOI: 10.3389/fenrg.2022.966422. [Simulation case study on a 3-pack / 15-cell system; SOH balancing → 52.9% simulated cycle-life improvement vs. conventional active SOC balancing.]

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[18] "Failure Investigation of $LiFePO_4$ Cells under Overcharge Conditions" (2012). *Journal of The Electrochemical Society*. [Fe^{2+} dissolution, Li plating and electrolyte oxidation cascade under LFP overcharge; A123 18650 cells.]

[21] EV Parts India (2025). "Cell Balancing Techniques in Lithium Battery BMS: Passive vs. Active Approaches." [Industry documentation — not peer-reviewed; passive threshold typically 30-50 mV; active trigger threshold 20-30 mV on commercial systems.]

[22] EM Battery Systems (2023). "Cell Balancing: How Active and Passive Processes Work in BMS." embatterysystems.com. [Manufacturer documentation — not peer-reviewed; states that "balancing to ±1 mV is not practical" for passive systems.]

[23] DALY BMS Technical Documentation (2025). "Active Balancing BMS vs Passive BMS: Which Do You Need?" dalybms.com. [Manufacturer documentation; active balancing trigger threshold default 20-30 mV; balancing ceases at threshold.]

[24] Cell Saviors Technical Guide (2024). "How to Force a BMS to Balance Lithium-Ion Batteries." cellsaviors.com. [Industry documentation; best commercial systems reach 3-5 mV configurable thresholds; low-cost systems 10-30 mV.]

[25] Orion BMS Utility Manual (2024). "Cell Balancing Description." orionbms.com. [Manufacturer documentation; recommended maximum delta voltage 10 mV for LFP; lower settings cause control instability in the balancing algorithm.]

[26] Pinto, J.G.O., Miranda, J.P.D., Barros, L.A.M. & Afonso, J.A. (2025). "Development of Modular BMS Topology with Active Cell Balancing." *Batteries*, 11(11), 421. MDPI. DOI: 10.3390/batteries11110421. [Prototype study of a 22-cell modular active-balancing BMS; reduced a deliberately large 1.18 V initial differential to ~0.47 V in 2 h 30 min. Cited as a worked example of active-balancing dynamics, not as a performance ceiling of the field.]

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